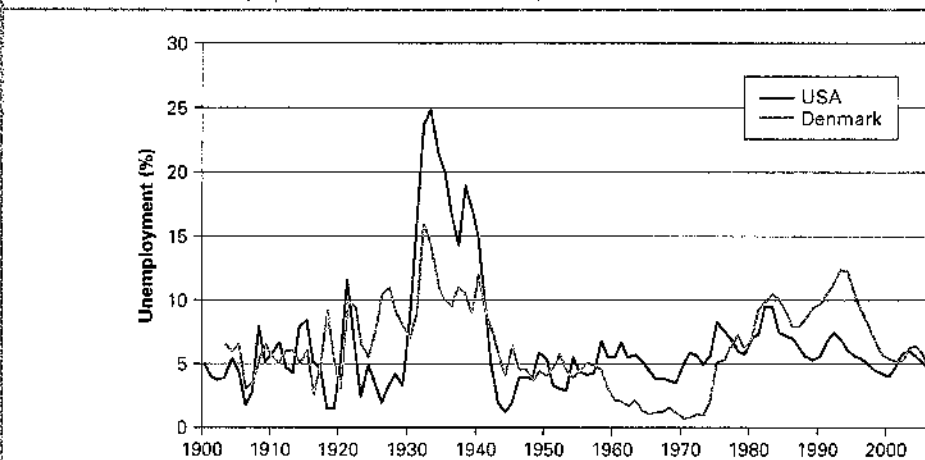


Some facts and introductory theory about unemployment

In the growth models studied in the previous parts of this book, adjustments in the real wage ensured that labour demand was always equal to labour supply, because markets were assumed to be perfectly competitive. Literally speaking, we assumed that there was no unemployment at any point in time. In the real world there is always at least some unemployment.

Figure 10.1 shows average annual rates of unemployment for the USA and Denmark over a long historical period. As we explained in Chapter 1, the movements in the rate of unemployment can be viewed as consisting of two components: a *long-run trend* which could be described as a tendency towards a constant unemployment rate around 5–7 per cent over very long periods, and *cyclical fluctuations* around the trend. The fluctuations are sometimes large, and perhaps so large that one would really not call them fluctuations. Occasionally unemployment has gone to very high levels in both countries. For instance, in the 1930s the rate of unemployment went up to 25 per cent in the USA and above 15 per cent

FIGURE 10.1 Rates of unemployment in the USA and Denmark, 1900–2006



Sources: R.B. Mitchell, *International Historical Statistics*, Macmillan, 1998; Bureau of Labor Statistics (US); S.A. Hansen, *Økonomisk Vækst i Danmark*, Bind 2, 1914–83, Akademisk Forlag, 1984; Statistics Denmark (Statistisk Tiårsoversigt).

in Denmark, and over the 1970s and early 1980s, unemployment rose to rates of around or above 10 per cent in both countries.¹

The fact that the long-run 'gravity' rate of unemployment is not zero but is rather in the range of 5–7 per cent is not a problem for the growth theory presented in this book. We can simply reinterpret the L_t appearing in the growth models as total employment when the labour supply is used at a normal, or natural, rate. Our growth models were just silent about what determines the natural rate of utilization.

For society it is important how high the long-run rate of unemployment is. It makes a difference if the average rate of utilization of the most important economic resource, labour, is 96 or 92 per cent. This chapter and the two next are focused on the forces that determine the long-run or so-called natural rate of unemployment.

10.1 The social cost of unemployment

High unemployment is a major economic and social problem and a concern of economic policy makers mainly for two reasons. At the individual level, when unemployment is high many persons will have to go through long spells of joblessness. They therefore suffer substantial income losses that may force them and their families to move from their homes, change the children's schools, take a cut in their standard of living, etc. Although unemployment insurance is available in most countries, it does not insure a person's income fully, and it covers only a limited period.² In addition to the loss of income, longer periods of involuntary unemployment may inflict serious psychological costs on the jobless, because an unemployed person may feel excluded from society and therefore lose self-esteem. Unemployment thus gives rise to serious individual problems.

At the level of society, unemployment is a social waste, because income that could have been earned to the benefit of the individual as well as society is forgone. The income that unemployed persons would have earned, had they had jobs, would have been taxed, and the tax revenue could have been used to the benefit of other citizens. When unemployment is high, the associated loss of tax revenues will be substantial. Thus the income loss of an unemployed person is partly suffered by the unemployed himself and partly by other members of society. In any case it is a waste of resources to have people who are both qualified and willing to work at the going wage rates, unable to do so.

Economists explain the social waste associated with involuntary unemployment in more detail as follows. Let the marginal product of (a particular type of) labour be MPL , that is, MPL is the increase in production that will arise if one more worker is put to work, or the decrease that will result if one individual is taken out of work. We think of the marginal product of labour as being decreasing in employment because of the presence of fixed factors, e.g. capital or land, or because the most productive workers are hired first.

The marginal product of labour must be held up against the marginal rate of substitution between consumption and leisure, MRS , also referred to as the marginal disutility of work. A worker's MRS is the amount of goods needed to compensate him for the leisure and the value of the home production he must give up when taking a job. Typically different workers will have a different MRS . A person will obviously want to work if offered a real wage above the person's MRS , so the total labour supply induced by a given real wage is the number of workers having an MRS below that wage rate.

If a worker's marginal product is larger than his or her MRS , then a social loss occurs if he or she does not work, since putting him or her to work gives an additional amount of

1. The data underlying Fig. 10.1 are not completely comparable between the two countries. However, for both countries the rates represent all unemployed as a percentage of all workers rather than all *insured* unemployed as a percentage of all insured workers.

2. In Chapter 19, where we discuss the costs of the cyclical part of unemployment, we will go into more detail with the reasons why unemployment insurance typically cannot and does not fully cover the income losses arising from unemployment.

output sufficient to compensate the worker and to leave a surplus that can be shared with other members of society (through taxation) to the benefit of everybody. A situation where $MRS < MPL$ thus implies inefficiency in the use of labour resources.

Now, if there is involuntary unemployment and firms maximize profits, there will indeed be jobless workers with marginal products exceeding their disutilities of work. An involuntarily unemployed person would like to work at the going real wage rate, W/P , so this wage rate must be greater than his or her disutility of work, $MRS < W/P$. Profit maximization implies that the last worker hired has a marginal product at least as great as the going real wage rate, that is, $W/P \leq MPL$, since otherwise the firm could increase its profits by firing the marginal worker. On the plausible assumption that the marginal product of labour decreases *gradually* with employment, there will be unemployed workers with marginal products close to that of the last worker hired. In other words, there will be unemployed workers for whom the condition $W/P \leq MPL$ is (very close to being) fulfilled. Collecting what we have found, if there is involuntary unemployment, then there must be unemployed workers for whom $MRS < W/P \leq MPL$, implying $MRS < MPL$. Hence their joblessness involves a social loss.³

For these reasons it is of great interest to know what determines the rate of structural unemployment and what policy measures could bring the natural rate of unemployment down from, say, 8 to 4 per cent. These are exactly the questions to which we now turn. This chapter lays the basis for answering these questions by going through some empirical regularities concerning unemployment and by presenting some introductory theories. The next two chapters will present two main theories of structural unemployment.

10.2 The rate of unemployment

Labour force, employment and unemployment

The rate of unemployment is a stock variable that can be measured at a given point in time, say on a specific day. Let the total number of persons who *would like to work* on day t be L_t . This is the labour force, which can be viewed as the size of the population of working age, P_t , times the participation rate, π_t , that is, $L_t = \pi_t P_t$. One can further subdivide into various groups such as women and men, $L_t = \pi_t^w P_t^w + \pi_t^m P_t^m$, where π_t^w and π_t^m are the participation rates of women and men, respectively. Total employment on day t , E_t , is the number of persons actually having a job on that day. The number of unemployed is then simply $U_t = L_t - E_t$, and the rate of unemployment is

$$u_t \equiv \frac{U_t}{L_t} = \frac{L_t - E_t}{L_t} = 1 - \frac{E_t}{L_t} \equiv 1 - e_t,$$

where we have denoted the rate of employment by e_t . Table 10.1 shows an estimate of the labour forces in the USA and Denmark for the month of January 2006, and it also shows the decomposition of the labour forces into employed and unemployed.

The rate of unemployment contains an element that is not as readily measurable as GDP, employment, inflation, etc., namely the number of persons, L_t , who *would like to have a job*. Out of these the E_t employed actually have a job, so we can infer that indeed these people want to work. The U_t unemployed, however, are persons who would like to work and yet do not do so. How do we measure the number of such people?

One way is to limit attention to those covered by unemployment insurance and to assume that individuals who plan to be active in the labour market will want to insure themselves against unemployment. One can then let L_t be the total number of people holding unemployment insurance, and let E_t be the number of insured people who have

3. An exercise will ask you to restate the explanation of the social cost of unemployment using a standard diagram.

TABLE 10.1 Labour market data, January 2006

(in 1000 persons)	USA	Denmark
Population in working ages*		
– Women	97,341	1,772
– Men	94,850	1,804
Participation rates		
– Women	89%	76%
– Men	81%	84%
Labour force		
– Women	66,951	1,348
– Men	76,797	1,515
Total labour force (L)	143,748	2,863
Employed (E)	136,320	2,755
Unemployed ($U = L - E$)	7,428	108
Rate of unemployment (U/L)	5.2%	3.9%

Note: *Population of working age is defined as those between 16 and 64 years of age in the USA, and between 16 and 66 years of age in Denmark.

Sources: Bureau of Labor Statistics (US) and Statistics Denmark (Statistik Tilsynet).

a job. Another way is to let L_t be the sum of all those who have a job (E_t) and all those who are registered at some kind of (public) institution as looking for a job but not having one (U_t). A third way is to let L_t consist of all persons who either have a job (E_t) or receive unemployment benefits or social assistance benefits motivated by joblessness (U_t). A fourth way is to estimate the number of unemployed by survey techniques asking a representative sample of individuals whether they are available for work. The three latter ways are the most common, and all the data for unemployment you will see in this chapter are of these types.

Although the rate of unemployment by definition pertains to a specific day, the data for unemployment most often shown are annual (or quarterly or monthly) rates. Fig. 10.1, for instance, shows annual rates. The annual rate of unemployment is simply the average over all the working days in the year of the daily rates of unemployment as defined above.

The duration of unemployment

When you hear that the annual rate of unemployment was 12 per cent in Denmark in 1993, you might think that 12 per cent of the people in the labour force were unemployed during all of 1993. This was not the case. Indeed, many more people than 12 per cent of the labour force experienced some unemployment during 1993, but most of these people were unemployed for considerably less than all of the year. Table 10.2 shows that in Denmark in 1993, the fraction of people in the labour force who experienced some unemployment was 29 per cent, but 21 per cent of these people were unemployed for less than five weeks, 44 per cent were unemployed less than 14 weeks, while 64 per cent were unemployed for less than half the year. A considerable 36 per cent of those affected by unemployment, that is, 10.4 per cent of all people in the labour force, were unemployed for more than half the year.

In the USA the annual rate of unemployment was substantially lower than in Denmark in 1993, namely 7 per cent, and the fraction of people in the workforce who experienced some unemployment was 20 per cent. Of these people, 20 per cent were unemployed for more than half the year, implying that only 4 per cent of the people in the labour force were unemployed for more than half the year.

This illustrates a general fact that we will return to below. Higher overall unemployment tends to raise the share of long-term unemployment in total unemployment. The figures for 2006 also confirm this. In Denmark the annual rate of unemployment had come down to

TABLE 10.2 Duration of unemployment

1993			2006		
Percentage of people experiencing unemployment during the year, who were unemployed for	USA	Denmark	Percentage of people experiencing unemployment during the year, who were unemployed for	USA	Denmark
Less than 5 weeks	37	21	Less than 5 weeks	37	38
5–14 weeks	29	23	5–14 weeks	32	27
15–26 weeks	14	19	15–26 weeks	15	16
27 or more weeks	20	36	27 or more weeks	16	19
	100	100		100	100
Persons having experienced unemployment as a percentage of the labour force	20	29	Persons having experienced unemployment as a percentage of the labour force	16	17
Unemployment rate, %	7	12	Unemployment rate %	5	4

Sources: Bureau of Labor Statistics (US) and Statistics Denmark (Statistik Tiansoversigt).

4 per cent, and now only 19 per cent of those affected by unemployment were unemployed for more than half the year, as opposed to 36 per cent in 1993 when the overall unemployment rate was 12 per cent.

If no people were ever unemployed for more than, say, one month during a year, the individual loss would not be so severe. However, those who are unemployed for long periods may be badly hurt by unemployment. A close positive relationship between the overall rate of unemployment and the rate of long-term unemployment provides one of the main reasons for trying to avoid high unemployment in general.

The Beveridge curve

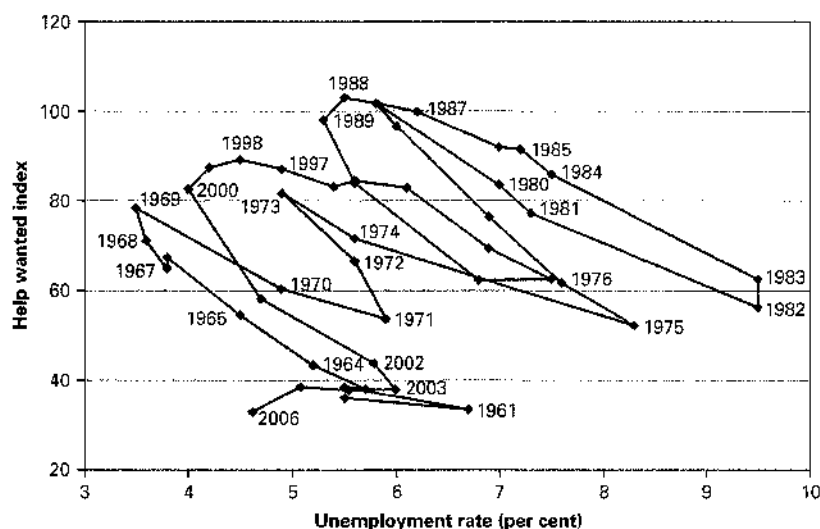
If there is (involuntary) unemployment, can there be vacant jobs at the same time? At first, one would perhaps say no. If there are unemployed people truly seeking jobs, they should take the vacant jobs until there are no more vacancies. If there is unemployment there should be no vacancies, and if there are vacancies there should be no unemployment. This would correspond to a perfect matching in the labour market between people who want jobs and the employers who offer jobs. However, the matching is never that perfect.

Just as one can define a rate of unemployment at a given time, one can define a rate of vacancies. Let J_t be the total number of jobs offered by employers on day t , and let E_t be total employment on that day. The number of vacancies is then $V_t = J_t - E_t$, and the vacancy rate is $v_t \equiv V_t/J_t$.

To estimate vacancy rates we must be able to measure the number of vacant jobs, V_t . This involves even more practical problems than measuring the number of unemployed people, U_t , but one can try in different ways. In the USA a 'help wanted' index is published. This measures the number of job advertisements in major US newspapers. We may take this index as an imperfect measure of the vacancy rate. If the measurement error is the same at different dates, a time series for the help wanted index should give a correct indication of the direction in which the 'true' vacancy rate moves.

The perfect matching between 'people wanting jobs' and 'jobs wanting people' described above would imply that in a diagram with u_t along the horizontal axis and v_t along the vertical axis, only points (u_t, v_t) on one of the two axes would be observed. Fig. 10.2 shows how the points are situated for the US over the period 1960–2006, if one uses the help wanted index as a proxy for the vacancy rate.

FIGURE 10.2 A Beveridge curve for USA, 1960–2006



Source: Bureau of Labor Statistics and The Conference Board.

The relationship between u_t and v_t is called a Beveridge curve. The fact that the curve does not just follow the axes is an indication of 'mismatch' in the labour markets. The further out to the northeast the curve is situated, the more severe are the problems of mismatch. The position of the Beveridge curve can thus be seen as an indicator of structural problems in the labour market. This is exactly why the Beveridge curve is useful. If the position of the curve shifts over time, this indicates shifts in the matching efficiency in the labour market. In Fig. 10.2 the points for the period from 1960 to 1969, for instance, are located on a nice downward-sloping curve. Then there is a shift outwards. The points for 1979 to 1988 seem again to be nicely situated on a downward-sloping curve, but this curve has shifted considerably to the north-east compared to the 1960–69 curve. This indicates that during the decade 1979 to 1988 people and jobs had much greater problems finding each other than during the decade 1960 to 1969. One would expect this to be part of the reason why there was more unemployment in the latter of these decades. From 1988 to 2006 there seems to have been an almost continually ongoing improvement of the basic structures of the American labour market.

10.3 Stylized facts about unemployment

We mentioned that the rate of unemployment contains an element that is not easy to measure, namely the number of people who do not have a job but *would have liked to have one*. For this reason there is some disagreement among economists about how much to trust statistics on unemployment rates. Should these be included among the time series that theories are held up against, or should one only include series of 'truly measurable' variables? Yet most economists believe that one can safely trust reported statistics of unemployment to give at least *some* indication of the extent of involuntary unemployment.

To see why, we may start by asking if there is a way to test if those who report that they are involuntarily out of work would actually take a job at the going wages if they were offered one. Assume that due to 'good times' there is a sudden increase in the demand for labour. If this reduces unemployment without driving up real wages too much, it can be taken as a sign that those who reported being unemployed actually did want jobs at the going wages, since they indeed took jobs at these wages when they had the chance. In other

words, if we observe a clear relationship between increases in GDP and decreases in the rate of unemployment and, at the same time, do not observe that real wages increase sharply as soon as GDP goes up, then this suggests that the officially reported rates of unemployment indeed contain true information.

In Chapter 13 we return to the empirical facts of the business cycle. One fact reported is that real wages are not strongly correlated with GDP over the cycle, and in particular there is no strong tendency that increases in GDP are associated with increases in real wages. At the same time there is a clear tendency that increases in GDP are accompanied by decreases in reported unemployment, as shown in Fig. 10.3 for the USA and Denmark.

The relationship illustrated by Fig. 10.3 is both important and useful. It shows, for instance, that at a constant rate of unemployment, economic growth in the USA is around 3 per cent per year while for Denmark the corresponding intersection is just above 2 per cent (the difference being mainly due to a higher growth rate of the labour force in the USA). A one-time increase in the growth rate of GDP of 3 percentage points reduces unemployment by around 2 percentage points in Denmark and by slightly less in the USA (the difference again due to higher labour force growth in the USA). The relationship reported in Fig. 10.3 has a name: Okun's Law.

STYLIZED FACT 1

(Okun's Law). There is a tight negative relationship between the rate of growth in GDP and the absolute change in the rate of unemployment.

The rate of unemployment is a fraction and by definition it has to stay between 0 and 1. Therefore it cannot increase by a certain percentage each year over many years, since this would eventually make it larger than 1. It could have an increasing trend in other ways, for example it could tend to increase from one level to another higher level. Figure 10.4 (as well as Fig. 10.1) indicates that nothing like that is true. The figure uses the same data as Fig. 10.1 and shows the annual absolute change in the rate of unemployment for the USA and Denmark. Over time spans of 100 years, increases and decreases seem to offset each other so that there is no upward or downward long-run tendency.

STYLIZED FACT 2

There is no upwards or downwards trend in the rate of unemployment in the (very) long run.

Note, however, that over quite substantial periods there can be a tendency for unemployment to rise or fall. Consider Fig. 10.1 again and assume that for the USA you only had the data from the early 1940s to the early 1980s, a period of about 40 years. For this long period there seems to be a sustained upward movement in the rate of unemployment. Something similar is true for Denmark for the period from the early 1970s to the early 1990s.

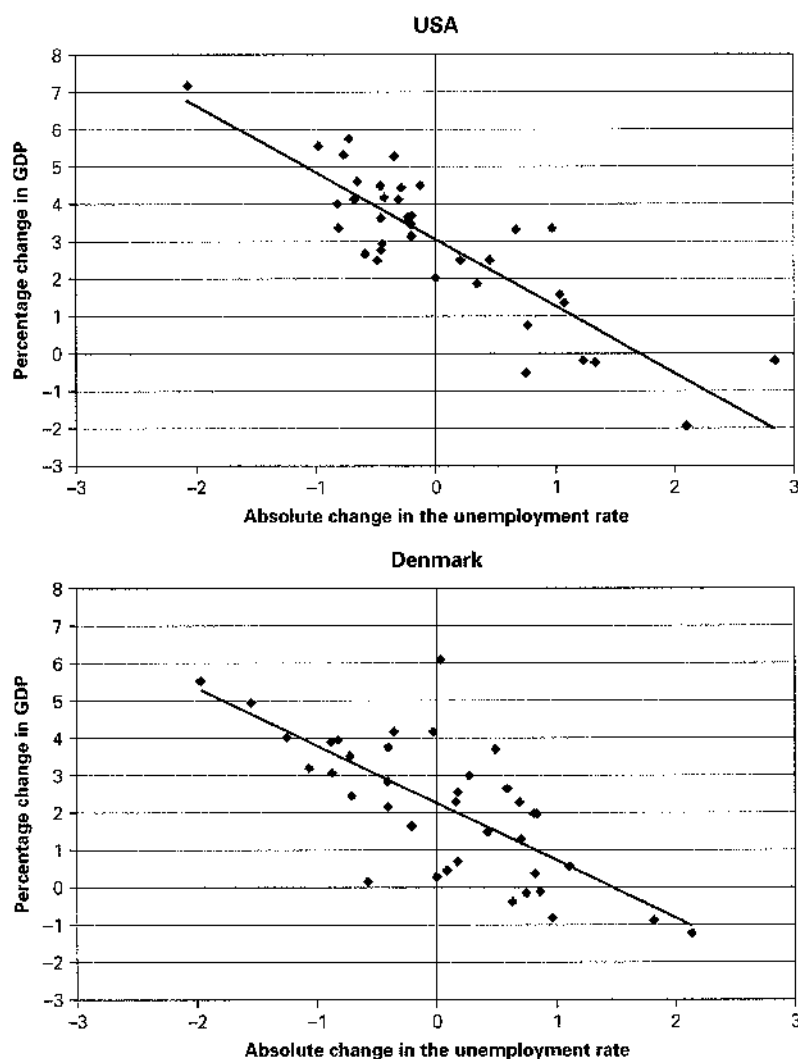
Figures 10.1 and 10.4 also bring evidence of another fact which is well known:

STYLIZED FACT 3

There is a lot of variability in the rate of unemployment in the short run.

A fact that is perhaps not so directly visible from Figs 10.1 and 10.4 is the persistence in the rate of unemployment: if the rate of unemployment is high in a specific year, then

FIGURE 10.3 Okun's Law, 1971–2008



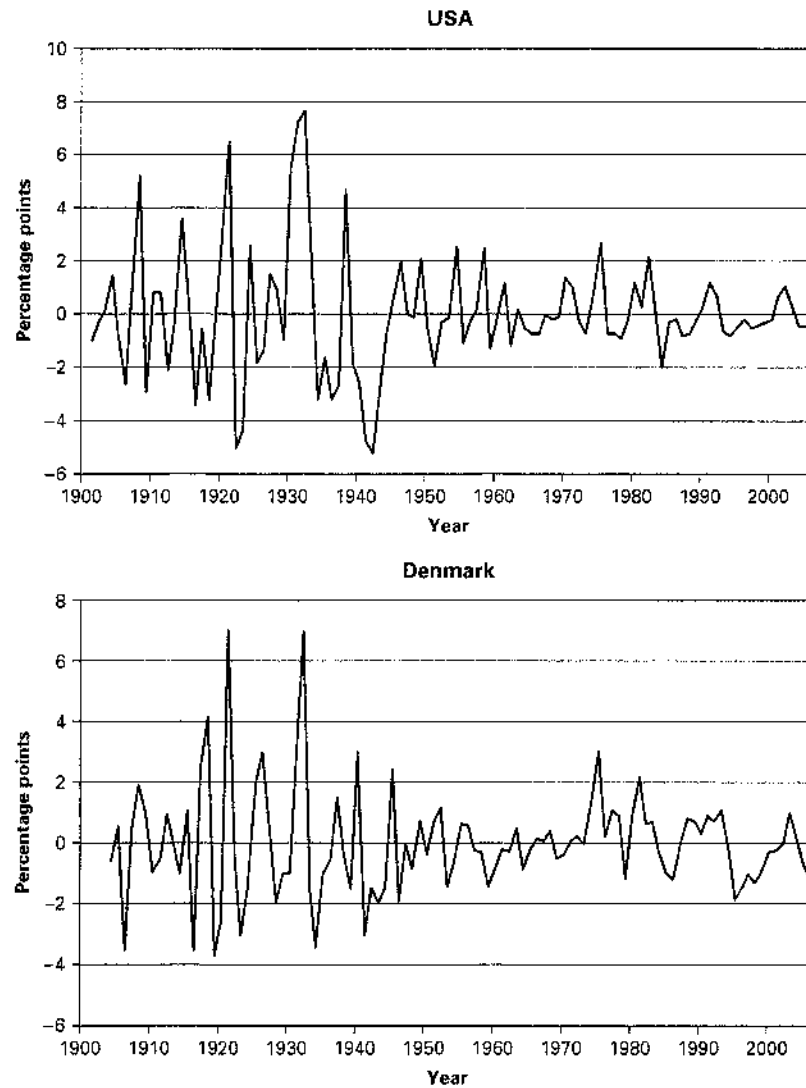
Note: The scatterplots show the annual percentage change in GDP against the annual absolute change in the rate of unemployment over the period from 1971 to 2008. Each dot corresponds to a single year. For the statistically proficient, the regression line for the USA has a slope of -1.8 , $R^2 = 0.73$ and $t = -9.8$, while the line for Denmark has a slope of -1.5 , $R^2 = 0.52$ and $t = -6.3$.

Source: OECD, Economic Outlook.

it was probably also high the year before (and will be the year after). We can measure this persistence by the coefficient of correlation between the current unemployment rate and its own past values. The coefficient of correlation between two time series, u_t and x_t , running through $t = 1, \dots, T$ and with averages $\bar{u} = (1/T)\sum_{t=1}^T u_t$ and $\bar{x} = (1/T)\sum_{t=1}^T x_t$ is:

$$\rho \equiv \frac{\sum_{t=1}^T (u_t - \bar{u})(x_t - \bar{x})}{\sqrt{\sum_{t=1}^T (x_t - \bar{x})^2} \cdot \sqrt{\sum_{t=1}^T (u_t - \bar{u})^2}}.$$

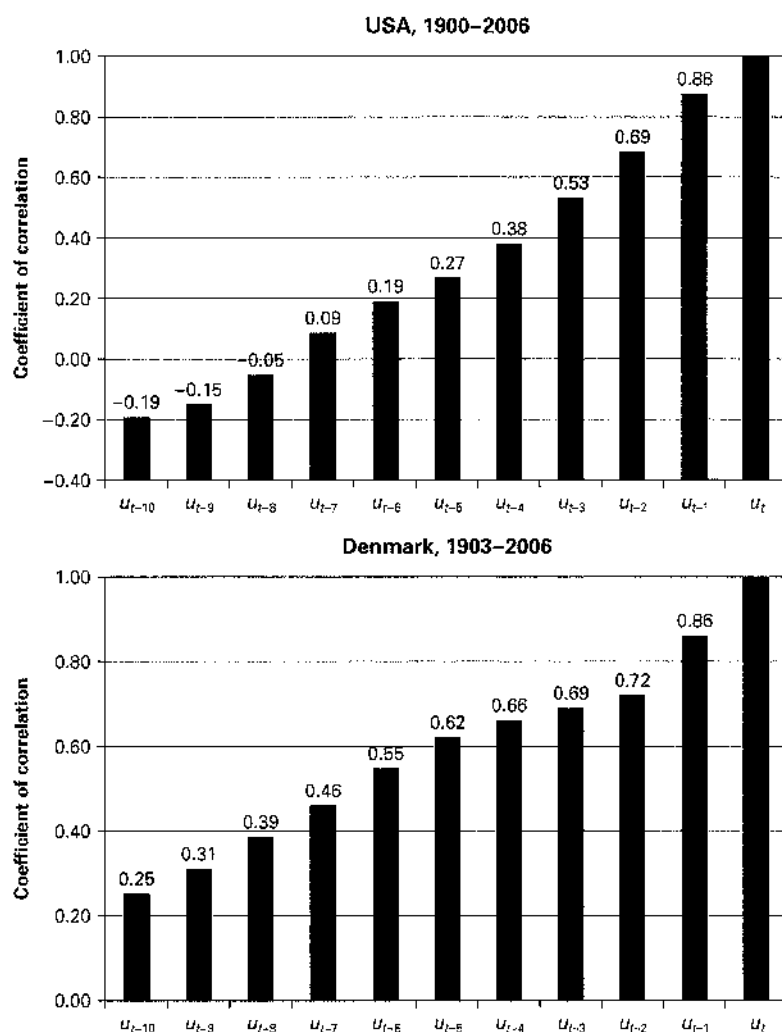
It should be visible from the formula that the coefficient ρ measures the degree of (linear) covariation between the two series, that is, roughly the degree to which one series is above

FIGURE 10.4 Annual absolute change in the rate of unemployment

Source: As Fig. 10.1.

average when the other is above average. One can show that $-1 \leq \rho \leq 1$, and $\rho = 1$ and $\rho = -1$ correspond to complete linear covariation along a straight line, $u_t = a + bx_t$, with positive and negative b , respectively. If u_t is the annual rate of unemployment and one lets $x_t = u_{t-1}$, and computes ρ for the two series u_t and x_t (for $t = 2, \dots, T$), one gets the coefficient of correlation between the rate of unemployment in one year and the year before. One can then let $x_t = u_{t-2}$ to get the correlation with unemployment two years before, etc. Figure 10.5 shows such coefficients of correlation in unemployment for the USA and Denmark.

There is a clear positive correlation between the current rate of unemployment and the rates of unemployment in several previous years, and the degree of correlation decreases as one goes back in time. The tendency for high unemployment in one year to imply high unemployment in several previous and succeeding years is what we mean by the persistence in unemployment.

FIGURE 10.5 Correlation in unemployment over time

Note: The columns illustrate the coefficient of correlation between the unemployment rate at time t and time $t-k$. Thus, the column u_{t-3} shows the degree of correlation between the unemployment rate and the unemployment rate three years before.

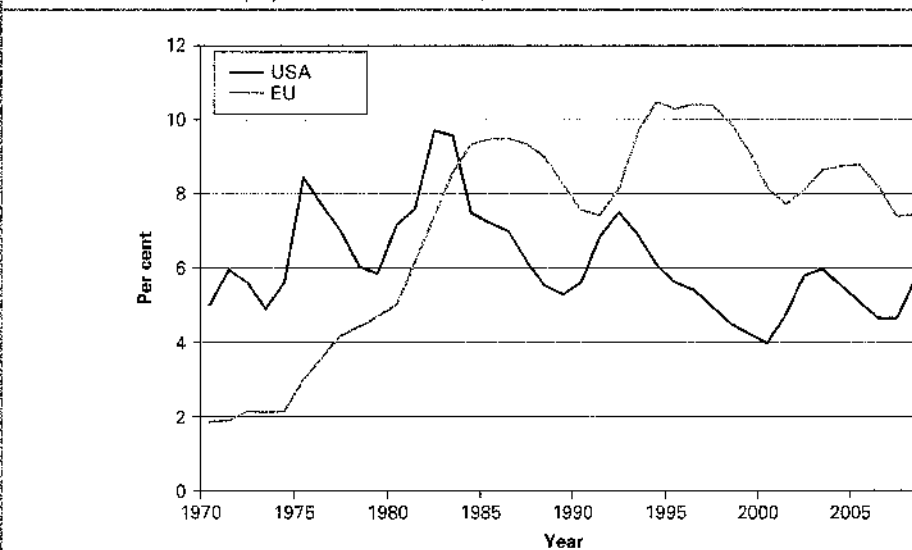
Source: As Fig. 10.1.

STYLIZED FACT 4

There is a lot of persistence in annual rates of unemployment.

In connection with Table 10.2 we examined the relationship between overall unemployment and long-term unemployment. We can compute the annual rate of long-term unemployment over several years, defining this rate as, for example, the percentage of all people in the labour force who were unemployed for more than half of the year. We can then plot the rate of long-term unemployment against the overall rate of unemployment. This is done for the USA and for Denmark in Fig. 10.6.

The relationship is evidently tight and positive, with a relatively steep slope. This motivates:

FIGURE 10.7 Rates of unemployment in the EU and USA, 1970–2008

Sources: OECD, Main Economic Indicators.

Does unemployment tend to be of equal size in different regions of the world? Perhaps it does in the very long run, as suggested by Fig. 10.1, but over substantial periods there can be considerable differences between major regions, as shown in Fig. 10.7. For a long time until the early 1980s, unemployment in Europe was lower than in America, but since then joblessness has been substantially higher in the EU than in the USA.

STYLIZED FACT 6

There can be large differences in unemployment across geographical areas for long periods of time.

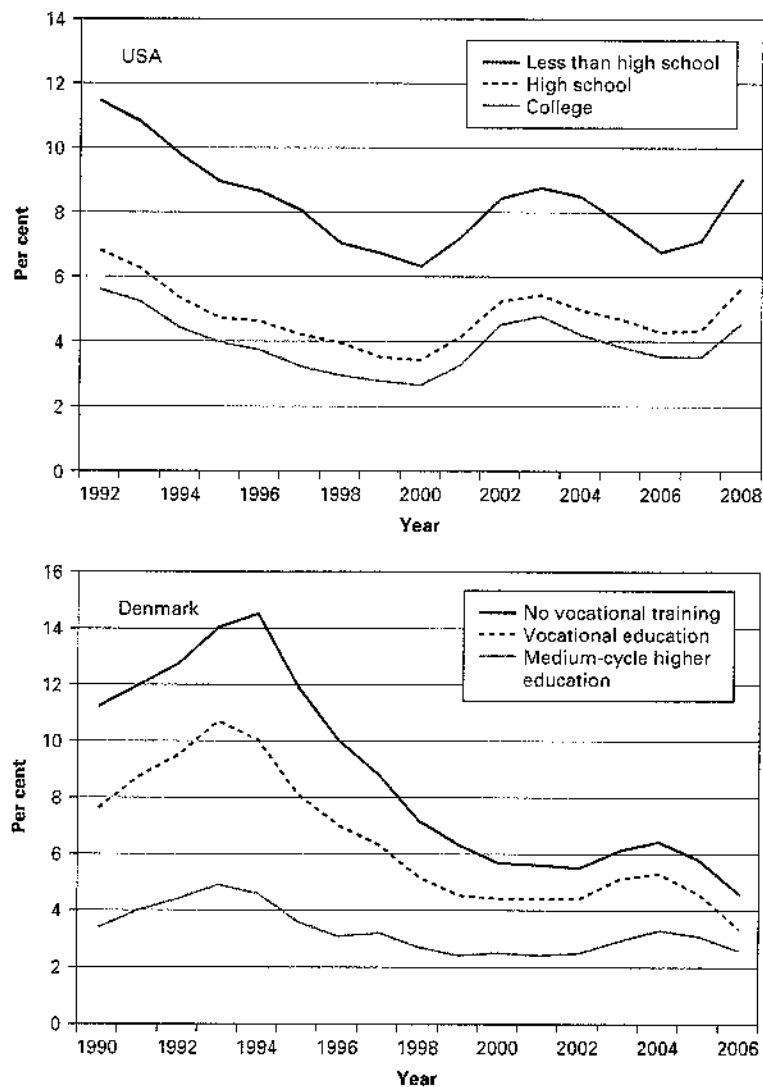
Other systematic variations are also important. Fig. 10.8 shows rates of unemployment across educational groups for the USA and Denmark. Although different educational categorizations have been used for the two countries, the general picture is clear:

STYLIZED FACT 7

There are considerable and long-lived differences in rates of unemployment across educational groups with a broad tendency for higher education to mean lower unemployment.

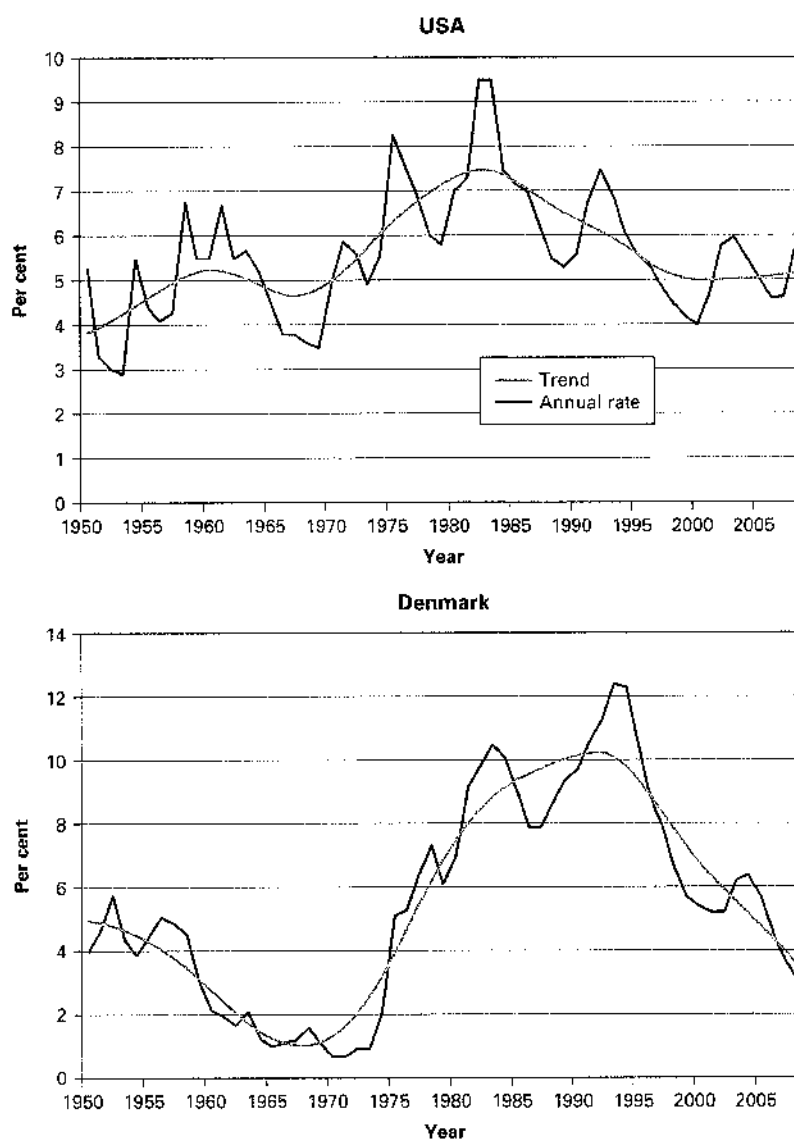
Variations across other categories are of importance in different connections, e.g. across race or sex. However, a relatively high rate of unemployment for a particular section of the population may, at least partly, reflect that this group contains relatively many unskilled people compared to the general population.

The final 'law' we will focus on here is an important one, and the remainder of this chapter is more or less centred around it. Figure 10.9 shows in two panels annual rates of unemployment for the USA and Denmark for the postwar period from 1950 to 2007. For each country a trend constructed by the so-called 'HP filter' that you will learn about in Chapter 13 is shown. Fig. 10.9 is a close-up of the more recent part of Fig. 10.1 and Fig. 1.2.

FIGURE 10.8 Unemployment by educational attainment

Sources: Bureau of Labor Statistics (US) and Statistics Denmark (Statistik Tårsoversigt).

Consider the upper panel for the USA. Even when the rate of unemployment is lowest, there is still a considerable amount of unemployment with annual rates normally above 4 per cent. Furthermore, the annual rate of unemployment fluctuates around a 'gravity level' that may change over time, but is normally not too far from 5–6 per cent. Also, for Denmark there seems to be a lower bound on the rate of unemployment and fluctuations around a gravity level, but the latter changes considerably more over time than for the USA. The period in the 1960s and early 1970s, when the unemployment rate in Denmark went far below 2 per cent, showed many signs that this was not a sustainable or equilibrium situation: inflation rose sharply. Accounting for this, the figures indicate that even in the best of times the lowest possible rates of unemployment are considerable, perhaps around 3–4 per cent, and gravity rates of unemployment are higher. Figure 10.9 gives a clear indication that the gravity or natural rate of unemployment can shift over time and differ between countries (and regions).

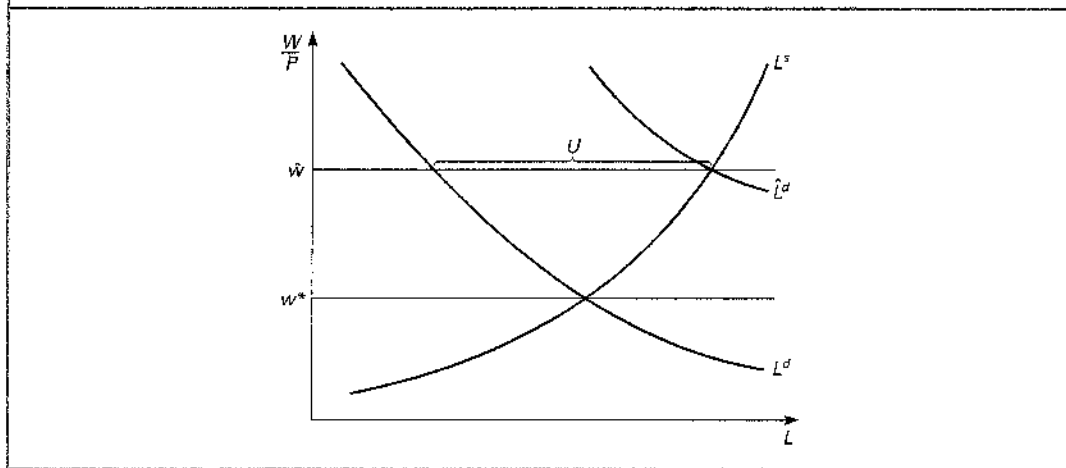
FIGURE 10.9 Annual rates of unemployment and their trends in USA and Denmark, 1950–2008

Source: OECD, Economic Outlook.

STYLIZED FACT 8

When rates of unemployment are at their lowest, there is still a substantial amount of unemployment, seemingly around 3–4 per cent, and the natural unemployment rate that the annual rates fluctuate around is higher. The natural rate of unemployment can shift over time and be different in different regions and countries.

FIGURE 10.10 A labour market



10.4 Short-run cyclical and long-run structural unemployment

This section links the various types of unemployment, short-run cyclical and long-run structural, to the different types of wage and price rigidity that economists think are associated with them. The section will therefore contain some repetition of material from Chapter 1.

Unemployment and excess supply of labour

Figure 10.10 illustrates a situation of unemployment in a labour market (disregarding mismatching). The figure assumes that the *individual* suppliers of labour take the nominal wage rate, W , as well as the nominal price level, P , as given. The labour supply curve, L^s , is drawn as an increasing function of the real wage rate, W/P , but nothing in our arguments would change if the labour supply curve were vertical. The firms that demand labour are also assumed to take the nominal wage rate as given, and the demand curve for labour, L^d , is decreasing in the real wage rate.

If this labour market were perfectly competitive, the market forces (working on the nominal wage rate, W) would imply a real wage at the level w^* , where supply equals demand and there is no unemployment. A situation of unemployment, as illustrated in the figure, means that the real wage rate is at some level $\bar{w}$ at which the supply of labour exceeds the demand for labour by the amount U .

Should the excess supply of labour not imply downward pressure on the nominal wage rate, forcing the real wage down to w^* , thus eliminating unemployment? This is a key question that any theory of unemployment has to face. Let us first just state two trivial, but nevertheless important, and highly related points in connection to this question:

- 1 In a labour market characterized by full competitive wage flexibility, there can be no unemployment, since the nominal wage rate adjusts immediately to generate a real wage that equates the competitive (wage-taking) supply of and demand for labour.
- 2 For unemployment to be possible, there must be factors preventing the (real) wage from adjusting to a level that would equate the competitive supply of labour and the competitive demand for labour.

Wage rigidity in the short run and in the long run

If we define 'wage rigidity' as any factor that prevents a full, competitive equilibration of the labour market, we can also summarize the two above statements like this: *for unemployment*

to exist there must be some wage rigidity. Usually economists distinguish between two types of wage rigidity and two associated types of unemployment.

Short-run nominal wage rigidity and cyclical unemployment

If a situation like the one illustrated by Fig. 10.10 prevails (for some time) for the mere reason that it takes time for the nominal wage rate to adjust to circumstances, we say that a short-run nominal wage rigidity is at work. The part of unemployment caused by short-run nominal wage rigidity we call short-run, or cyclical, unemployment.

Short-run nominal wage rigidity can take different forms. One possibility is that the adjustments of the nominal wage rate are really governed by competitive market forces, that is, by discrepancies between wage-taking supply and demand, but it simply takes some time for an excess supply of labour to bring about a full downward adjustment of the nominal wage rate. The reason is that an excess supply of labour has to be there for some time before it can invoke a downward pressure on the nominal wage rate. There will therefore be a period where the real wage rate can be at a level such as $\hat{w}$ in Fig. 10.10 simply because the nominal wage adjustments that should bring it down *have not yet occurred*. We would expect cyclical unemployment caused by such a nominal wage rigidity to be eliminated by market forces alone in the longer run when nominal wages have had time to adjust fully. However, if the economy is hit by new shocks all the time, the nominal wage rate may be on a never-ending job of adjustment, and during this continual adjustment, there will occasionally be unemployment and occasionally excess demand for labour or 'overemployment'.

Another possibility that we introduced in Chapter 1 is that the nominal wage rate is really set by an economic agent, let us say a trade union that organizes the wage-taking individual workers. As we explained in Chapter 1, if the nominal wage rate is set optimally at the outset and there are small 'menu costs' associated with nominal wage adjustments, it will be optimal for the trade union *not* to react to shocks as long as these are not too large and of a temporary nature.

The real wage, W/P , may be flexible under short-run nominal wage rigidity because the nominal price level, P , may adjust in response to conditions in the product markets. The implied adjustments in the real wage should not be expected to eliminate unemployment since they do not come in response to circumstances in the labour market.

The dynamics of cyclical unemployment arising from short-run nominal wage rigidity are one of the main subjects of Book Two and will not be dealt with further here.

Long-run real wage rigidity and structural unemployment

Consider again the situation in Fig. 10.10 where the wage-taking labour supply exceeds employment, and assume that the real wage rate has already fully adjusted *in accordance with the real wage flexibility that the economy possesses*. In that case we say that a long-run real wage rigidity is at work, and the part of unemployment that is caused by long-run real wage rigidity we call long-run, or structural, or natural unemployment.

Note that this is different from short-run nominal wage rigidity. It is not that the nominal wage rate has not *yet* adjusted to circumstances or that wage-setting agents have *preferred* not to react to temporary and opposing shocks. Here we assume (artificially) that the economy has not been exposed to new shocks for a long time so that all relative prices have fully adjusted to circumstances in accordance with the economy's long-run price and wage flexibility, and yet excess supply of labour prevails. We should, of course, be interested in the exact reasons why an excess supply of labour would not imply a downward adjustment of the real wage rate in the long run. We briefly consider such reasons below, while Chapters 11 and 12 deal in depth with two particular and important reasons for real wage rigidity and their macroeconomic implications.

The division of all unemployment into short-run and long-run unemployment is closely related to the Stylized fact 8 above, and to the nature of the unemployment fluctuations illustrated in Figs 10.1 and 10.9. From time to time there is a sudden increase in the rate of unemployment, for instance by two percentage points over just one year. It is natural to see

this as caused by a negative shock to which the economy reacts not only with lower production, but also (temporarily) with higher unemployment because of short-run nominal wage rigidity. This can explain the fluctuations in the rate of unemployment, but not the trend level of 5–7 per cent that the unemployment rate fluctuates around. Over time spans such as the more than 50 years in Fig. 10.9, or the more than 100 years in Fig. 10.1, we cannot possibly view the economy as being all the time reacting to negative shocks. Sometimes the shocks must have been mainly positive, sometimes the labour market must have been close to equilibrium where wage rates are adjusted in accordance with the economy's wage flexibility. The permanent part of unemployment must therefore be due to long-run real wage rigidity, i.e., to the fact that the price and wage flexibility of the economy is simply not such that it tends to eliminate *all* unemployment, not even after a long time without new shocks. The box at the end of this section illustrates our categorization of unemployment and sums up other issues as well.

One of the main reasons why economists find it useful to distinguish between the fluctuating or cyclical part of unemployment associated with short-run nominal wage rigidity, and the permanent or 'natural' part of unemployment associated with long-run real wage rigidity, is that the kind of policies that can be expected to work in fighting the two types of unemployment are different. Long-run unemployment can only be affected by policy measures that have an impact on the price and wage flexibility of the economy. We call such policies *structural policies*. One example could be reforms of the unemployment insurance system designed to make the unemployed exert a stronger downward pressure on wage rates, but as we saw in Chapter 1, a policy which increases the degree of product market competition can also help to reduce structural unemployment. Short-run unemployment, on the other hand, can be affected by *stabilization policies*, such as traditional fiscal and monetary policy.

Causes of long-run structural unemployment

For long-run real wage rigidity and structural unemployment to exist, there must be some reason why unemployment does not impose a downward pressure on wages. The remaining part of this chapter, and Chapters 11 and 12, are focused on this issue.

Why should the presence of unemployed people not give rise to downward pressure on wage rates, not even in the long run? Let us go through some possible reasons.

Abuse

One potential reason is that some people registered as unemployed may not really be looking for jobs but may only be interested in collecting the publicly subsidized unemployment benefits. Hence they do not contribute any downward pressure on wages. There is hardly any doubt that such abuse occurs to some extent, but we have argued above that probably most of registered unemployment is not of this type, and in the following we will disregard it.

Frictional unemployment

Another reason is that some of the unemployed have already found a job, but have not yet started in it. Usually people are not hired immediately, but, say, from the first day of the next month. There will then be a period of time during which they are unemployed, and involuntarily so, but they do not look for jobs since they already have a job starting a few weeks later. Due to the natural development of the economy there will all the time be some people who lose their jobs. For instance, as time goes by the products of some firms become outdated, the firms must close, and the employees lose their jobs. Even at times when these people do not have difficulty finding new jobs, there will be some unemployment due to a number of people being on their way from one job to another, since the transition takes some time for technical and formal reasons beyond people's control. We call this *frictional* unemployment, and frictional unemployment is one part of structural unemployment. It is

of interest to try to estimate how much structural unemployment is frictional, and we will return to that in the next section. The conclusion will be that probably only a minor part of long-run unemployment is frictional. Hence there is still a lot of structural unemployment to explain, and indeed some economists view the non-frictional part of long-run unemployment as the 'true' structural unemployment.

Search unemployment

If the people who lose their jobs spend some time before taking a new job, not for reasons beyond their control, but as a matter of choice (they may turn down job offers for some time), we call the associated unemployment *search* unemployment. Just like frictional unemployment, search unemployment cannot be expected to exert downward pressure on wages. The two types of unemployment are closely related, differing only in the reason why the unemployed turn down job offers. In the case of frictional unemployment it is because they have already accepted a job offer. In the case of search unemployment it is because the unemployed choose to wait, or search, for better job offers. Who says that the first offer received is the best one the unemployed worker can get? Perhaps it is wise to spend some time searching. Search unemployment is thus closely related to uncertainty about available job offers: it is caused by imperfect information about job possibilities. Furthermore, search unemployment is voluntary.

The level of search unemployment depends on a basic trade-off for the unemployed person. On the cost side, longer search for a new job implies a larger income loss during the waiting time. On the benefit side, it implies a higher expected future income from the day employment starts, assuming that more search ultimately leads to a better job offer. The more important the benefit side is relative to the cost side, the more (voluntary) search unemployment there will be. Generous unemployment benefits help to keep the cost side down; they work as a subsidy to search. Therefore, lower benefits would tend to reduce search unemployment, and this is one example of a structural policy that could be used to fight a type of structural unemployment.

Should policy makers try to reduce search unemployment as much as possible? No, there are some positive aspects of search unemployment. It is in the general interest of society that the qualifications of the labour force are used optimally, that each individual is employed where his or her productivity is the highest, since this means higher incomes. The search by individuals may work to ensure that people are placed in jobs where they earn well because they have high productivity. So, for society there is also a basic trade-off related to search unemployment. When people are 'separated' from their jobs, we want them to search for some time in order for them to find new jobs where they have high productivity, but we do not want them to search for too long, because then the output loss during the search period will outweigh the productivity gain from further search.

The job for policy makers is thus not to reduce search unemployment to a minimum, but to create incentives through the tax and benefit systems for an *appropriate level of search unemployment*.

Search unemployment and structural policies intended to affect it are important issues in macroeconomics. However, the formal theory of search unemployment is quite complex, and for that reason we will not have (much) more to say about search unemployment in this book.

Efficiency wages

The efficiency wage theory of the coexistence of unemployment and fully adjusted wages represents a minimum departure from the fully competitive model of wage adjustment. Since it is the subject of Chapter 11, we will be brief about it here. Just note that there is a hidden assumption behind a supply and demand diagram like Fig. 10.10, or more precisely behind the labour demand curve in it. The labour demand curve is given by the decreasing marginal product of labour, since labour is demanded up to the point where the marginal cost of an additional unit of labour – the real wage – equals the marginal revenue product. However, behind this construction is the tacit assumption that the wage paid to the workers does not

affect their productivity. In Chapter 11 we argue that if we keep all other characteristics of the competitive labour market, but add the feature that productivity may depend positively on real wages, then the traditional labour demand curve 'disappears', and firms may choose a combination of a real wage and a number of people demanded inside the labour supply curve. The result may well be unemployment without any tendency for falling wages. Wages fail to adjust because firms do not *want* lower wages, and unemployment is involuntary since individual unemployed workers would really like to work at the going wage rate.

Trade unions

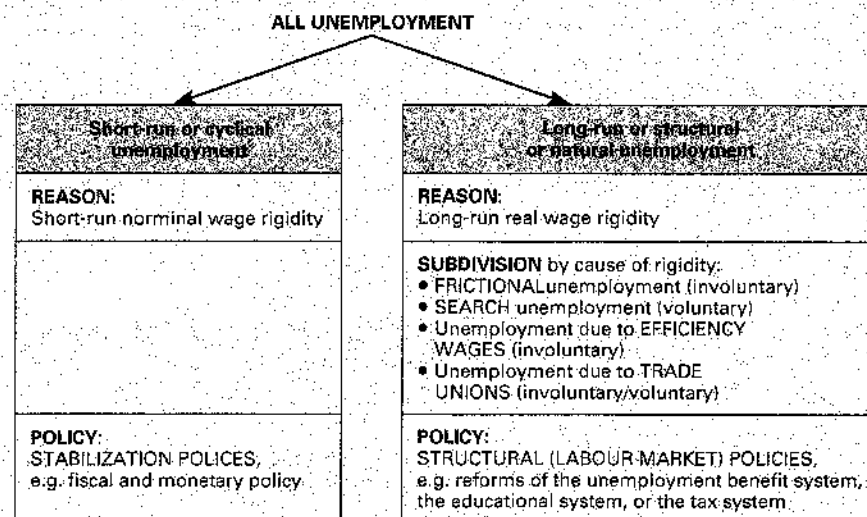
Unionization of the labour market represents a more 'brutal', but realistic and therefore important, departure from the competitive model. It is simply assumed that the supply side of the labour market (and sometimes also the demand side) is not atomistic, since the workers are organized in a trade union that takes care of their interests. The trade union sets the wage rate and the labour market therefore does not work competitively, but more like a monopoly market. As we shall see in Chapter 12, this may well imply that the wage rate chosen by the trade union will be so high that the wage-taking labour supply of all the workers on the labour market exceeds the labour demand of the firms. Hence there is unemployment, but there is no tendency for falling wages, since the wage rate is what those who set it want it to be. The unemployment will be involuntary at the individual level, since the unemployed workers would really like to work at the wage rate set by the union. At the collective level, however, the unemployment has something voluntary about it, since it is due to the fact that workers have decided to organize to exert influence on the wage rate and, having obtained market power through a union, let this union *choose* a wage rate that is so high that unemployment results.

In this book we focus on efficiency wage setting and trade union behaviour to explain why real wages do not adjust to eliminate long-run unemployment. To motivate this focus, we will now argue that a large part of long-run unemployment cannot be explained by frictional or search unemployment.

UNEMPLOYMENT TAXONOMY



The figure below presents compactly our categorization of unemployment as well as the causes of the different types of unemployment and the economic policies intended to fight them.



10.5 How large is frictional and search unemployment?

Let us now return to the simple formula that relates the number of unemployed workers to the labour force and the number of employed workers, say in a specific month, $U_t = L_t - E_t$. For simplicity assume that the labour force is constant, $L_t = L$ in all months t , so that the formula is $U_t = L - E_t$. How can the number of unemployed workers possibly change from month t to month $t + 1$? In exactly two ways.

Some people who are employed in month t may lose or quit their jobs from t to $t + 1$, causing an increase in unemployment. We denote by s_t the 'separation rate' of month t , that is, the fraction of the employed in period t who lose their jobs from t to $t + 1$. Then the increase in the number of unemployed coming from job separation can be written as $s_t E_t$. Some people who are unemployed in month t may find a job from t to $t + 1$, which gives a reduction in unemployment. If we denote the 'job finding rate' of month t by f_t , this reduction in unemployment is $f_t U_t$. The flow diagram in Fig. 10.11 shows the two ways the number of unemployed workers can change.

The total change in the number of unemployed workers from month t to $t + 1$ is thus:

$$U_{t+1} - U_t = s_t E_t - f_t U_t.$$

If we divide both sides by L and remember that in any month the rate of unemployment is $u_t = U_t/L$, we get:

$$u_{t+1} - u_t = s_t \frac{E_t}{L} - f_t u_t.$$

Now use $E_t = L - U_t$ to get:

$$u_{t+1} = (1 - s_t - f_t)u_t + s_t.$$

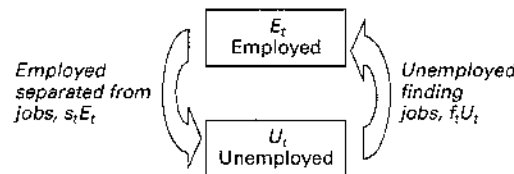
Given the evolution of s_t and f_t and an initial rate of unemployment, this equation describes how the rate of unemployment evolves over time. This law has nothing in particular to do with frictional or search unemployment. It is simply an identity that has to hold for unemployment in general (given a constant labour force).

We will now focus on a 'stationary environment' where both s_t and f_t are constant over time, that is, $s_t = s$ and $f_t = f$ in all months, t . You may find this assumption strange. Shouldn't the possibility of finding a job, the job finding rate f_t , depend on how much unemployment there is? We make this assumption deliberately since we want to focus entirely on frictional and search unemployment. We do not want it to be particularly difficult to find a job. We want to find out how much unemployment there will be if the only reasons for unemployment are that when people lose their job for some reason, they do not start in new jobs immediately, either for purely technical/practical/formal reasons, or because they decide to search for an attractive job offer. The dynamics of the rate of unemployment now simply become:

$$u_{t+1} = (1 - s - f)u_t + s. \quad (1)$$

Here we should think of s as a relatively small positive fraction, perhaps around 0.02, corresponding to 2 per cent of the employed workers losing their jobs each month.⁴ If everyone who loses a job starts in a new one 'the first day of the next month', the job finding

4. In their article 'The Cyclical Behavior of the Gross Flows of US Workers', *Brooking Papers on Economic Activity*, 2, 1990, Olivier Blanchard and Peter Diamond find a monthly job separation rate for the USA over the period January 1968 to May 1986 of 2.9 per cent. In the period from 1968 to 1986 there was an increasing trend in the rate of unemployment in the USA (see Fig. 10.9), so the separation rate under the stationary conditions we want to focus on here is probably somewhat smaller.

FIGURE 10.11 The flows in and out of unemployment

rate would be 1, but since voluntary job search will pull f down, we should probably think of f as a fraction that is somewhat smaller than 1, e.g. $f = 0.8$. This will imply that the parenthesis $(1 - s - f)$ in front of u_t in Eq. (1) is a positive number smaller than one, so the dynamics given by (1) are such that u_t converges monotonically to a constant level u (draw a diagram with u_{t+1} as a function of u_t and iterate) given by $u = (1 - s - f)u + s$, or:

$$u = \frac{s}{s + f}. \quad (2)$$

One can compute the u given by (2) for various values of s and f . From the above, a value 0.02 for s could be realistic. If the value of f is to be explained only by frictions or search, a value of 0.8 seems small. A large s and a small f tend to make the u in (2) large, so inserting $s = 0.02$ and $f = 0.8$ should give an indication of a large rate of frictional and search unemployment. With these parameters the computed u is around 2.4 per cent. Even if we send f all the way down to 0.5, which is surely low, the computed u will still be lower than 4 per cent (3.8 per cent to be precise).

We conclude that there seems to be important elements of structural unemployment that cannot be explained by frictions or search. There seems, therefore, to be good reasons to investigate other explanations of structural unemployment.

10.6 Summary

1. Unemployment implies individual costs in the form of income losses that cannot be fully insured as well as psychological stress. Unemployment implies a social cost as well. For an involuntarily unemployed worker the real value of the goods and services needed to compensate for the sacrifice of leisure (MRS) is smaller than the going real wage. When firms maximize profits, the real wage equals or is smaller than the marginal product of labour (MPL). Hence involuntary unemployment implies $MPL > MRS$. This means that a social gain could be achieved if some unemployed workers got jobs, since the extra output generated by additional employment would more than suffice to compensate the newly employed workers. Because unemployment is a social waste, it is a main concern of economic policy makers.
2. The rate of unemployment is the fraction of unemployed in the total labour force, measured at a given point in time. The annual rate of unemployment is the simple average of the rates of unemployment in each day or month of the year. A large number of workers are affected by unemployment in each year, typically around 20 per cent of the labour force in countries like the USA and Denmark, but only some of the unemployed are out of work all of the year, whereas the rest have unemployment spells of varying but shorter duration.

3. The long-term unemployed during a given year are those who were unemployed more than a given fraction of the year, for example more than half of the year. Long-term unemployment is the part of overall unemployment carried by the long-term unemployed. It is mainly the long-term part of unemployment that causes serious individual costs, but for the social cost the overall (average) unemployment rate is also important.
4. Unemployment and vacancies can coexist because the labour market's match between 'people wanting jobs' and 'jobs wanting people' is not perfect. The Beveridge curve is a plot over time of the vacancy rate against the unemployment rate and gives an indication of the degree of mismatch in the labour market. For given structural mismatch problems, the Beveridge curve should be a stable decreasing relationship. If it shifts inwards or outwards, it can be an indication of decreased or increased structural problems in the labour market, respectively.
5. The empirical evidence shows that (i) there is a clear negative relationship between the rate of growth in GDP and the change in the rate of unemployment, (ii) there is no upward or downward trend in the rate of unemployment in the very long run, (iii) there is a lot of variability in the rate of unemployment in the short run, (iv) there is a lot of persistence in annual rates of unemployment, and (v) long-term unemployment varies positively and more than proportionally with overall unemployment.
6. When annual rates of unemployment are at their lowest, there is still a substantial amount of unemployment, seemingly around 4 per cent, and annual unemployment rates seem to fluctuate around a natural unemployment rate of about 5–7 per cent. The natural rate of unemployment seems to shift over time and can be different in different regions of the world.
7. The presence of unemployment reflects that wage rigidities prevent the real wage from adjusting to equalize supplies and demands for labour. Economists find it useful to distinguish between short-run nominal wage rigidities, which are fundamental for explaining the cyclical fluctuations in unemployment, and long-run real wage rigidities, which cause the positive and constant trend level of structural unemployment. This part of the book is focused on the natural rate of unemployment and long-run real wage rigidities.
8. This chapter briefly presented the following four causes of long-run real wage rigidity and structural unemployment: (i) labour market frictions, (ii) job search, (iii) efficiency wages and (iv) market power of trade unions. The last two of these potential causes for structural unemployment are dealt with in the following two chapters.
9. Studying a model of the flows in and out of unemployment, we derived a simple formula linking the rate of unemployment to the separation rate (the fraction of employed who lose their jobs in a month) and the job finding rate (the fraction of unemployed finding a job from one month to the next). Assuming a stationary environment with constant separation and job finding rates, and assuming realistic values for these rates, we found that only a minor part of structural unemployment seems to be due to labour market frictions or search. This gives good reason to study the explanation for structural unemployment offered by the theories of efficiency wages and trade unions presented in our two next chapters.

10.7 Exercises

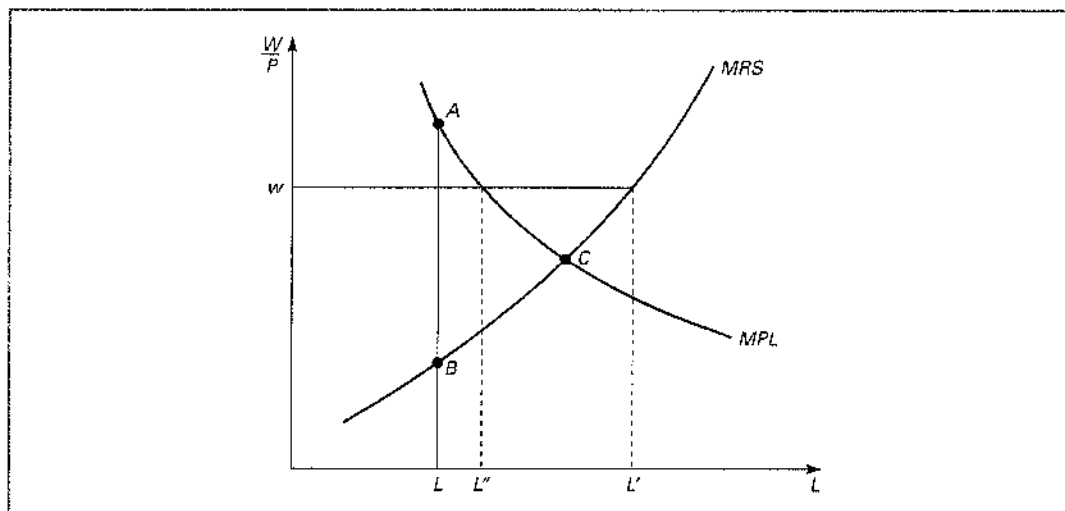
Exercise 1. The definition of unemployment

Exactly how is the rate of unemployment defined and measured in your own country according to the (most) official statistics? In particular, how is the number of unemployed counted? How is the labour force measured?

Exercise 2. The social cost of unemployment

This exercise asks you to explain the social cost of unemployment more or less as it was done in this chapter, but to do so using a standard diagram that you will see again in Chapter 19.

In the figure below, illustrating a particular labour market, the curve MPL depicts the marginal product of labour as this depends on the total number of workers employed. The curve is decreasing because of the presence of fixed factors or because of productivity differences among workers or both. For each worker there is a certain marginal rate of substitution of output for leisure, MRS , the amount of goods it takes to compensate the worker for the loss of leisure and home production resulting from having a job. The MRS curve in the figure shows how this marginal rate of substitution varies with employment when workers are lined up in order of increasing MRS .



1. Explain that if the individual workers are wage and price takers then the MRS curve is identical to the labour supply curve, such that if the going real wage rate is the w indicated, total labour supply should be the L' indicated, that is, the labour supply coming from workers with an MRS below w .
2. Explain that if the firms that buy labour in the market considered are wage and price takers then their labour demand curve is given directly by the MPL curve, such that the labour demand arising from a wage rate of w is the L'' indicated in the figure. Now assume that the firms take the nominal wage rate, W , as given, but, due to some kind of imperfect competition in the output market, profit maximization leads to the nominal price, P , being set as a mark-up over nominal marginal cost, that is, $P = m(W/MPL)$, where $m > 1$. Explain that to be in accordance with the profit-maximizing behaviour of firms, a combination of a real wage, w , and employment, L , must be such that at this combination, $W/P = MPL/m$, meaning that the labour demand curve is MPL/m . Indicate such a labour demand curve in the figure, and explain that under the circumstances described labour demand given w must be less than L'' .

3. Explain that if there is involuntary unemployment (some workers, who would like to work at the going real wage, are without jobs) then the prevailing combination of real wage, w , and employment, L , must be strictly to the left of the MRS (labour supply) curve. Then explain that if firms are wage takers and maximize profits, the combination of w and L must be (weakly) to the left of the MPL curve.
4. Explain that if there is involuntary unemployment and the firms maximize profits then at the prevailing level of employment one must have $MPL > MRS$. Explain why the situation involves a social loss. Assume that the combination of real wage and employment is as indicated by w and L in the figure, a situation of involuntary unemployment compatible with profit maximization. Explain that the area of the triangle ABC represents the total social loss in the situation. Assume that the combination is instead w and L'' , compatible with the firms being perfectly competitive, but still a situation of involuntary unemployment. Is there still a social loss and if so how large is it?
5. Return to the initial situation with real wage w and employment L , indicated in the figure. Assume that some marginal, hitherto unemployed workers are hired at the wage rate w , so that employment increases to a level above L , but below L'' . Indicate the social gain that this gives rise to as an area in the figure. Is there a way to make everybody enjoy part of the gain?

Exercise 3. The duration of unemployment

Assume that you are told that in some year the (average) unemployment rate was 8.3 per cent. Does this mean that 8.3 per cent of the people in the labour force were without jobs the whole year, or does it mean that all people in the labour force were without jobs for one month during the year? Or does it mean something else, and if so what does it (typically) mean? We said in this chapter that with respect to the individual costs of unemployment it is long-term unemployment that is most severe. What is meant by long-term unemployment and why is this a particularly bad kind of unemployment?

Exercise 4. Coefficients of correlation

Below is a table showing the rate of unemployment in the USA for the period from 1965 to 1975.

t	u_t
1965	4.51
1966	3.78
1967	3.84
1968	3.55
1969	3.51
1970	4.98
1971	5.95
1972	5.60
1973	4.88
1974	5.52
1975	8.47

Source: OECD Economic Outlook.

1. Find u_{t-1} by lagging the numbers in the table, such that, for instance, $u_{1966-1} = u_{1965} = 4.51$. There are only 10 pairwise observations between u_t and u_{t-1} . Why?

2. Compute the coefficient of correlation between u_t and u_{t-1} , using the formula for the coefficient of correlation, where $x_t \equiv u_{t-1}$:

$$\rho = \frac{\sum_{t=1}^T (u_t - \bar{u})(x_t - \bar{x})}{\sqrt{\sum_{t=1}^T (u_t - \bar{u})^2} \cdot \sqrt{\sum_{t=1}^T (x_t - \bar{x})^2}},$$

and $\bar{u}$ and $\bar{x}$ are the averages, $\bar{u} = (1/T)\sum_{t=1}^T u_t$ and $\bar{x} = (1/T)\sum_{t=1}^T x_t$. In this case $\bar{u}$ and $\bar{x}$ are close to each other. Explain.

3. Next, compute the coefficient of correlation between u_t and u_{t+1} . How does it relate to the one between u_t and u_{t-1} ? Explain your findings.

Exercise 5. Types of unemployment: reasons and policies

The box on page 295 presented a certain categorization of unemployment according to type. Explain the underlying motivations for this categorization, theoretical (in terms of different types of rigidities) as well as empirical. Explain also the figure's categorization of policies meant to fight unemployment, in particular why these policies are different for different types of unemployment.

Exercise 6. The size of frictional unemployment

Assume that due to normal 'job rotation', 2 per cent of workers lose or quit their jobs every month. Assume further that the only reasons why workers who are separated from their jobs do not find new jobs immediately are traditional labour market frictions, so all unemployment is frictional, and that these frictions are such that all workers who become unemployed in one month start in their new jobs the first working day in the next month. How large is the rate of frictional unemployment under these circumstances? How large is it if 2.9 per cent of all workers are separated from their jobs each month and only 90 per cent of those who are unemployed in one month find jobs in the next month? Why do we believe that not all structural unemployment is simply frictional?